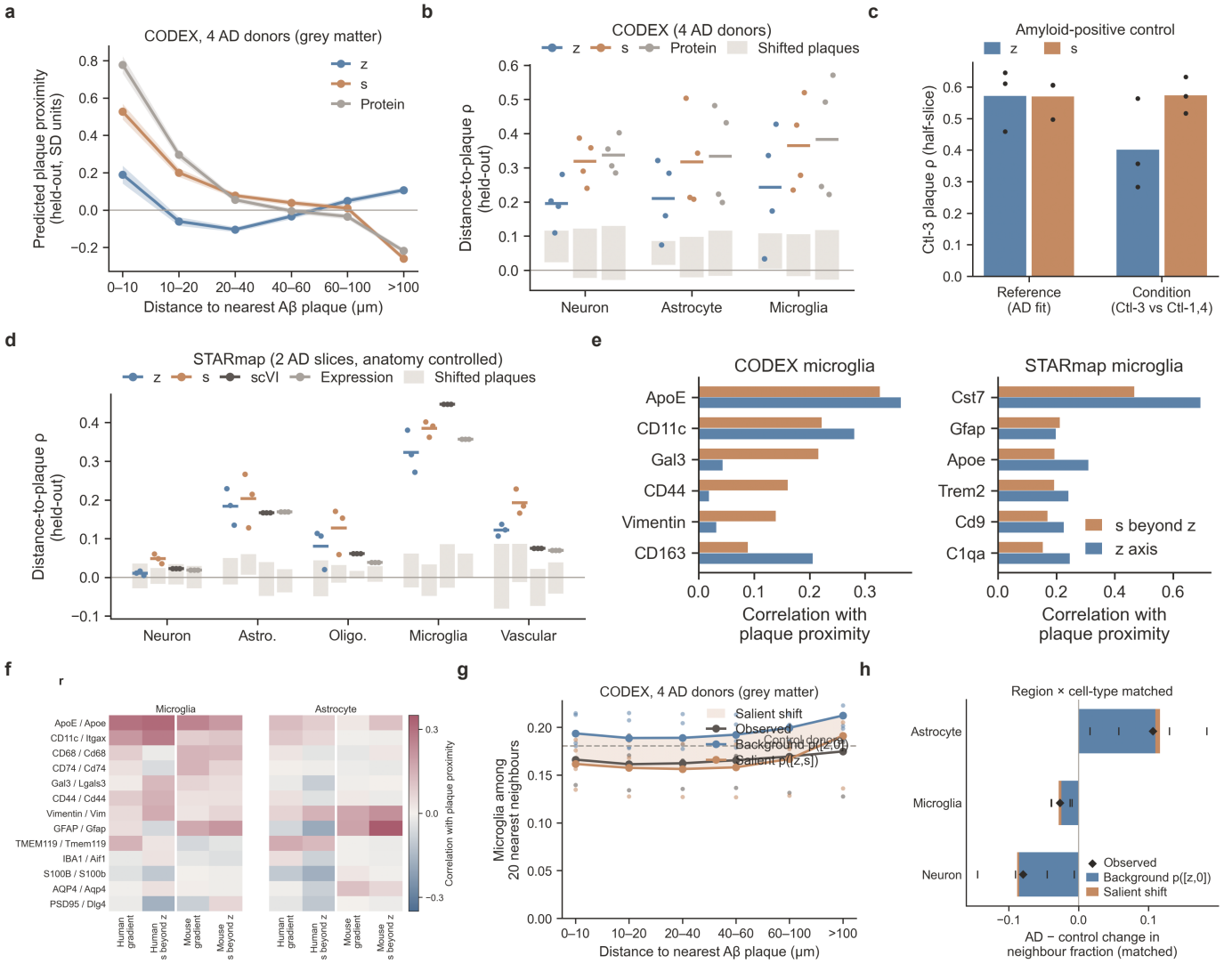

a
CODEX, 4 AD donors (grey matter)
z
s
Protein
Predicted plaque proximity (held-out, SD units)
0–10
10–20
20–40
40–60
60–100
>100
Distance to nearest Aβ plaque (μm)
b
CODEX (4 AD donors)
z
s
Protein
Shifted plaques
Distance-to-plaque ρ (held-out)
Neuron
Astrocyte
Microglia
c
Amyloid-positive control
z
s
Ctl-3 plaque ρ (half-slice)
Reference (AD fit)
Condition (Ctl-3 vs Ctl-1,4)
d
STARmap (2 AD slices, anatomy controlled)
z
s
scVI
Expression
Shifted plaques
Distance-to-plaque ρ (held-out)
Neuron
Astro.
Oligo.
Microglia
Vascular
e
CODEX microglia
ApoE
CD11c
Gal3
CD44
Vimentin
CD163
Correlation with plaque proximity
STARmap microglia
Cst7
Gfap
Apoe
Trem2
Cd9
C1qa
s beyond z
z axis
Correlation with plaque proximity
f
r
Microglia
Astrocyte
ApoE / Apoe
CD11c / Itgax
CD68 / Cd68
CD74 / Cd74
Gal3 / Lgals3
CD44 / Cd44
Vimentin / Vim
GFAP / Gfap
TMEM119 / Tmem119
IBA1 / Aif1
S100B / S100b
AQP4 / Aqp4
PSD95 / Dlg4
Correlation with plaque proximity
Human gradient
Human s beyond z
Mouse gradient
Mouse s beyond z
g
CODEX, 4 AD donors (grey matter)
Salient shift
Observed
Control domain
Background p([z,s])
Salient p([z,s])
Microglia among 20 nearest neighbours
Distance to nearest Aβ plaque (μm)
h
Region × cell-type matched
Astrocyte
Microglia
Neuron
Observed
Background p([z,0])
Salient shift
AD − control change in neighbour fraction (matched)